Proposed attack and detection algorithms

Claims

Threat Identification

Table 1 Threat Actor Specification

	Description	Reference
Motivations	Anti-Atom an anti-nuclear power organisation targeting UK nuclear powerplants. Their hope is this will cause the UK government to focus on alternative green means for power production. They think that the biproduct of nuclear power being radioactive waste is detrimental to the environment.	(Anon-a, 2024)
Intent	Their intentions are to reduce the availability and reliability of the powerplant leading to higher scrutiny and costs. This will be caused by shortening the lifespan of components and thus increasing repairs and inspections. This will play on the already inflated costs of nuclear powerplants, without alerting investigations with reactor trips or large deviations.	(Lawson, 2024)
Opportunity	Insider threat from a human actor which allows them to compromise an engineering workstation. As it is an insider threat they will have valid credentials to the system. Also detailed knowledge of the system and how it functions and responds to changes.	(Ayodeji, 2023)
Presence	Months of passive historian data collection to establish baselines and assess changes. Then minor changes over time within normal ranges to go unnoticed. No new software used just existing hosts and their software. Only visible due to subtle anomalies in write patterns.	
Tactics	Passive reconnaissance and initial access will be done by the insider with access to the systems with valid account credentials. Execution/ Persistence will come from scheduled task which will run the script at specific times only for certain durations to avoid detection. This script will inject false data for the steam generators. Discovery is from monitoring and polling the OPC UA nodes. With the final impact being a wear attack on the steam generators reducing availability and reliability.	(Anon-b, 2025)
Techniques	Manipulation of control by false data injection to the CTRL_SG1PressSetpoint and CTRL_SG2PressSetpoint setpoints. Whilst spoofing feedback so that the attack is not detected.	(Anon-c, 2025)
Procedures	Passive Reconnaissance. Initial access with valid credentials. Execution via scheduled task. False Data injection. Evidence suppression. Eventual wear attack.	

Target System – Steam Generators

Role of the Steam Generator

The steam generators convert the heat produced from the reactor core into steam. The steam then drives the turbines that generate the electricity. Ageing steam generators are a major factor that has a great impact on reliability and availability of a nuclear powerplant as they are vital to functionality and safety (Anon-d, 2011). Operating steam generators with high pressure or level conditions will accelerate fatigue and corrosion of components like heat exchanger tubes or feedwater nozzles and shells (Anon-d, 2011).

OPC UA Input Variables

CTRL_SG1PressSetpoint, CTRL_SG2PressSetpoint, CTRL_SG1LevelSetpoint, CTRL_SG2LevelSetpoint, RC1_PumpDiffPress, RC1_PumpSpeed, RC1_PumpFlow, RC1_PumpTemp, RC2_PumpDiffPress, RC2_PumpSpeed, RC2_PumpFlow, RC2_PumpTemp.

OPC UA Output Variables

SG1_InletWaterTemp, SG1_OutletSteamTemp, SG1_InletWaterFlow, SG1_OutletSteamFlow, SG1_WaterTemp, SG1_SteamTemp, SG1_Press, SG1_Level, SG2_InletWaterTemp, SG2_OutletSteamTemp, SG2_InletWaterFlow, SG2_OutletSteamFlow, SG2_WaterTemp, SG2_SteamTemp, SG2_Press, SG2_Level

Stealthy Attack Definition

Stealthy attacks are attacks that are purposefully built to remain hidden from the detection mechanisms that are in place whilst still having the desired impact. The definition of a stealthy attack depends on what intrusion detection system is in place; this section will focus on a physics-based system. The intrusion detection system generates predictions of expected outputs from the given inputs and a process model. An attacker will manipulate sensors or control signals giving false data, but the injected data will still follow the expected physical behaviour of the system (Urbina, 2016). When the anomaly detector compares the predicted outcome and computes a test statistic, the value will remain within normal bounds and will not exceed the alarm threshold.

Another type of approach is a stochastic linear system which defines both a stealthy and strictly stealthy attack. A stealthy attack must produce similar measured outputs according to the detectors predicted outputs thus keeping the residuals left over small and indistinguishable from noise. A strictly stealthy attack although not the focus would produce exactly zero residuals thus making it almost impossible to detect (Sui, 2020).

Both approaches follow similar logic and method for detection; the stealthy attack must follow the predicted physics of the system whilst not making any sudden and drastic changes to keep residuals small. As a result of this, stealthy attacks are usually a longer approach which occurs more subtly rather than an obvious quick failure.

Scope

Out of scope for my detection algorithms are during times of high noise for example changing the reactor setpoint as the high variance in the readings of features will make an attack indistinguishable. Attacks are also out of scope if the given normal for comparison occurred during an attack, as since the baseline is malicious then the attack will go unnoticed since no differences would be flagged.

Argument

Stealthy Attack Implementation

Attack Summary

The attack targets the steam generator pressure setpoints CTRL_SG1PressSetpoint and CTRL_SG2PressSetpoint from a compromised node injecting false data.

The stealthy drift attack function reads and stores the original value of the nodes then computes a drifting mean from the original value to that value plus the drift amount. This occurs over half the duration and while it is increasing it has a cosine oscillation to avoid a perfectly smooth ramp up and keep it looking normal. The modified setpoint is written at a fixed rate and for this I have chosen 10 Hz as it is on the lower end whilst still having the desired effect. In the second half the values drift back from the value plus drift back to the original value which stops any drastic jumps keeping it normal looking.

The mask attack function takes both the analogue and digital outputs (excluding the simulation time as it needs to constantly increase) and reads each node multiple times with 1 second in between reads. This is done to get a realistic set of values to write that includes variance to look normal. These values are replayed in a forward then backwards pattern at the same rate, so they do not jump and appear to be fluctuating normally. This also occurs for 3.25 times the duration to allow for the outputs affected by the attack to return to normal before they stop being masked. Once completed all nodes are reset to their original values.

The attack function that calls both above functions has two phases. Phase 1 increases both SG1 and SG2 pressure setpoints by the set drift amount. Phase 2 gradually decreases both the pressure setpoints by the negative drift amount. Both of which occur at the same time as the mask attack that repeats the outputs.

Potential Impact

Pressure cycling in the steam generators can lead on to greater wear on the steam generators components (Wang, 2026) this shortens components lifetimes. This also causes fluctuation in temperatures of the steam generators which causes thermal cycling (Anon-e, 2005) another cause of fatigue in critical components. These cycles don't only cause typical wear characteristics like cracking and thinning of components they can also influence the thermal performance of the generators lowering efficiency (Srikantiah, 1998). Aswell as cycling increasing the pressures it also lowers them after to help reduce the fuel consumption and offset the increase to attempt to hide the attack further (Anon-f, 2005). This helps it not get flagged by using more fuel then it should.

The effect on the power plant due to the above would be an increased probability of leaks which have both environmental and safety risks. More outages and maintenance will occur which affects availability and reliability. Both will cause a negative economic effect for the power plant as more money will have to be spent on repairs and outages.

Threat Actor Assumptions

This works under the assumptions that the attacker can reach the network where the OPC UA server is, to compromise the node and to run said attacks. This means they must possess valid credentials to access the site and to be able to get to a system that can allow this. The actor will need to have a good understanding of the system, so they know which nodes to attack and which to mask to avoid detection.

Detection Algorithm Implementation

Overall Strategy

My approach for a stealthy detection algorithm is two detectors that focus on different ideas to cover a larger range and complement each other effectively. This takes from the idea of stacking detection algorithms together to make them more effective and reduce false positives (Nguyen, 2022). One of the detectors is random forest that is trained on labelled data meaning it is supervised. To complement this well I use a Kalman and a CUSUM residual based detector as this is unsupervised. The idea behind this is that random forest captures behavioural patterns whereas Kalman and CUSUM can detect inconsistencies and drifts. I will also be polling all variables on the powerplant as inputs and outputs and giving them to the detection algorithms as this is widespread practice.

Kalman and CUSUM

The Kalman algorithm creates a 1-dimensional Kalman filter for each feature using the normal data. The filter learns the normal values that the feature typically has and how quickly it normally changes. Then the filter is run on the attack data which predicts what each variable should be based on the normal behaviour and compares them against the true value. Then it calculates the difference which is the residual and if that exceeds what is considered normal noise then it is flagged as an anomaly.

The CUSUM algorithm is ran on the absolute residuals that have been generated by the Kalman algorithm. Instead of focusing on large deviations CUSUM accumulates small deviations over time, making it well suited for detecting slow drift behaviour. Once the score exceeds a given per-feature threshold that is based on the normal behaviour then it is flagged as anomalous. This reduces false positives as it scales per feature.

I chose CUSUM as although it has a higher false positive rate, in cyber physical systems detection delay might be more important than potential false positives (Akowuah, 2021). It also covers most generated fail scenarios and has a low missed alarm rate (Wang, n.d.). I chose Kalman as it is proven to work well against both stealthy and un-stealthy attacks (Basiri, 2018) but it does have its draw backs as it doesn't work for all stealthy attacks and faulters for replay attacks (Ahmed, 2016). This lead on to me also implementing random forest to have an effective supervised machine learning model to back them up that does not work directly off residuals.

CUSUM has a per feature threshold allowance which is multiplied by twenty to allow for deviation without unnecessary flagging of slight changes. This works as it is cumulative so it can still detect multiple slight changes and scales per feature. Kalman has a NIS threshold for anomaly detection which I have set to 10 with a per feature noise variance and process noise variance. This works as it again scales per feature and does not flag on slight deviation which would be classed as normal behaviour for the specific feature.

Random Forest

The random forest algorithm (RF) is implemented by giving it pre-labelled normal and attack data sets as two csv files the normal labelled with zero and attack labelled with one. To train and test the model the datasets are combined into one and the feature set is built. The combined data set is split 70% training set and 30% test set to make sure its random but reproducible. Then a random forest model is created and trained on the training data and then evaluated on the test data using a classification report to get accuracy, recall and F1-scores for both attack and normal cases. The model is then retrained on the full data set to maximise learning before detection scores are generated. Lastly the fully trained model calculates per row attack probabilities for the attack data set. It does this by examining patterns across multiple features simultaneously so when relationships between variables is affected then it can detect that it is anomalous. This makes it a good pairing for statistical and model detectors.

I chose the RF algorithm as it is more accurate in comparison to other decision tree algorithms and linear regression (Khare, 2024). It was also commonly used as well as other machine learning algorithms and generally placed as the most effective (Teixeira, 2018) when dealing with not only variable changes but also network traffic. It also fits well into a multiple detection algorithm strategy well such as this (Anthi, 2021).

The default probability threshold of 0.5 is used to decide the attack vs the normal, whilst recording the raw probability for the graph. This allows for a fair split without flagging too many false positives as attacks, just by requiring a majority probability.

Evidence

Stealthy Attack Characterisation

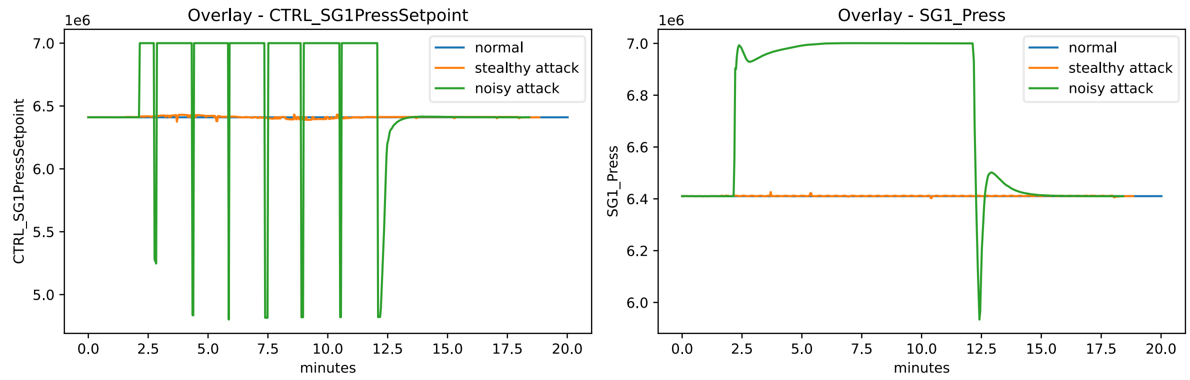

Figure 1 Stealthy Attack, Noisy Attack and Normal behaviour plotted against each other.

In the first graph in Figure 1, the CTRL_SG1PressSetpoint is shown which is the feature that the attack is targeting. The noisy attack (green line) overwrites the setpoint causing it to drastically change throughout. The stealthy drift attack (orange line), by comparison has a tiny adjustment which could easily be mistaken as normal noise. This shows that, even on the targeted setpoint the stealthy attack is nearly indistinguishable from normal behaviour (blue line) when compared to the noisy attack.

In the second graph in Figure 1, the SG1_Press is shown which is one of the direct outputs from the steam generator. The noisy attack (green line) causes it to drastically change throughout due to its effect on CTRL_SG1PressSetpoint. The stealthy attack (orange line) shows that the variable fluctuates enough to appear normal whilst concealing the effects of an attack. This demonstrates that even on the direct SG1 output, the stealthy attack remains almost indistinguishable from normal behaviour (blue line) due to it being masked effectively, unlike the noisy attack.

To conclude the stealthy attack is successful in making slight, normal looking tweaks that accumulate over time. These changes look negligible compared to the noisy attack, even for the targeted variable whilst masking changes in other variables. The attack would go unnoticed to operators although it would be heavy on network traffic.

Detection Algorithm Evaluation

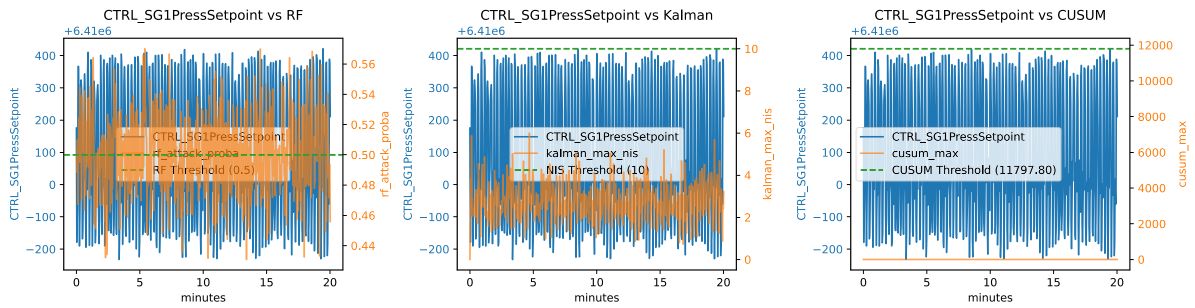

Figure 2 Normal Behaviour given as an attack to the detection algorithms.

I chose the SG1_PressSetpoint for all the graphs as it is the variable we are directly affecting and not masking. In Figure 2s first graph, the RF probability is plotted against the normal behaviour with the normal behaviour given as an attack. The attack probability remains consistently low and fluctuates around 0.5 not flagged as an attack. In Figure 2s second graph, the Kalman residual is plotted against the same. As you can see the residual remains below the threshold and remains low and random. The same behaviour is observed for CUSUM in the third graph, as it is driven by the Kalman residuals. The statistic remains below the anomaly threshold. These graphs show that none of the detection algorithms flag false positives when comparing normal behaviour.

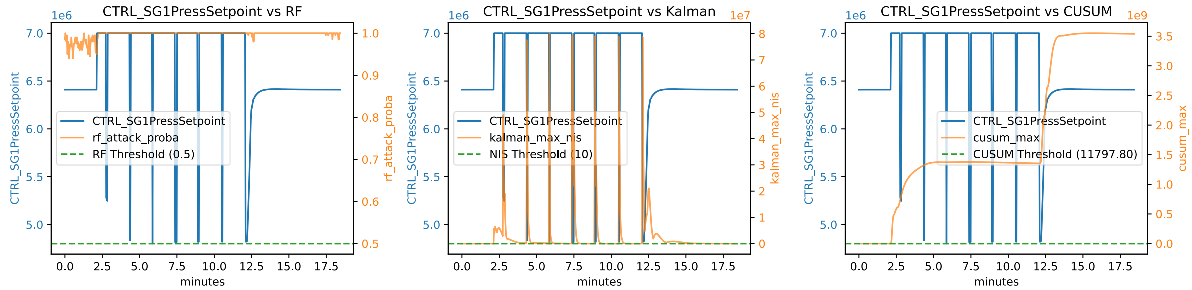

Figure 3 Noisy attack plotted against normal behaviour for the detection algorithms.

In the first graph in Figure 3, the RF probability is plotted against the normal behaviour with the noisy attack. The RF probability that it is an attack is at 100% for majority of the polled duration. This remains at 100% even after the attack has finished due to the attacks after-effects. In the third graph in Figure 3, the CUSUM residual is plotted against the same. As you can see the CUSUM residual spikes as soon as the attack occurs and remains elevated until after the attack stops. The same trend is visible for the Kalman residuals in the second graph. However, it flags fewer anomalous rows because it works off individual residuals rather than cumulative deviations. From all the graphs you can see that as soon as the noisy attack starts the three detectors trigger and either raise above the threshold or jump to 100% probability of an attack. This shows that all three detectors are effective for detecting noisy attacks.

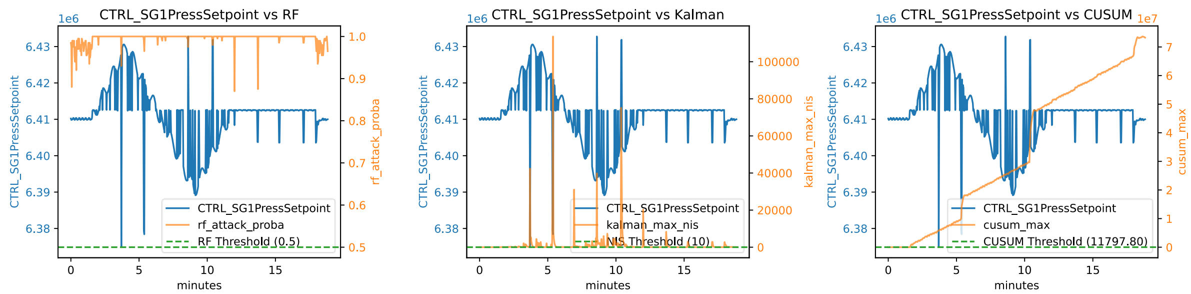

Figure 4 Stealthy attack plotted against the normal behaviour for the detection algorithms.

In Figure 4s first graph, the RF probability is plotted against the normal behaviour with the stealthy drift attack. It detects the attack with almost 100% certainty shortly after it starts. This is due to it tracking the relationships between correlated features. But it remains at 100% certainty for less rows than during the noisy attack. In Figure 4s third graph, the CUSUM residual is plotted against the same. As you can see the CUSUM residual increases much slower than the noisy attack and reaches a lower cumulative amount, but it is still over the detection threshold. This is the same in the second graph for Kalman, but it follows the curve of the overwrite attack peaking when the attack is at both its maximum and minimum points excluding anomalies. Both the CUSUM and Kalman responses are lower than those observed for the noisy attack, only peaking with anomalous readings during polling. From all the graphs you can see a delayed trigger for the attack with RF flagging first but Kalman and CUSUM taking longer to register and flag the attack. This shows that they can successfully detect the stealthy attack in normal conditions, but it is harder to detect and to flag than the noisy attack.

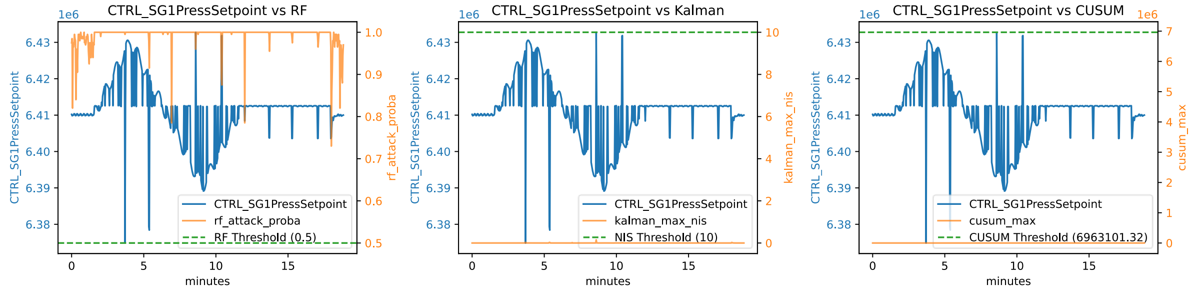

Figure 5 Stealth attacks plotted against a noisy normal baseline.

Then the algorithms were assessed under periods of transition. In the first 10 minutes of the normal the reactor setpoint is changed from 100 to 90 in 5 minutes then back to one hundred in the following 5 minutes. It remains at one hundred for the remainder of the 10 minutes. When using the detection algorithms against this as the normal baseline CUSUM and Kalman flag no anomalous rows. The RF algorithm detects the attack with near 100% certainty, but when we compare the transition normal against the regular normal it also registers it as an attack. This demonstrates that none of the detectors can reliably identify stealthy attacks during transitions, meaning that the normal baseline must be gathered at a stable state. This is shown in Figure 5 with the first graph showing the RF 100% probability plotted for the majority. Whereas the second and third graph both show negligible amount for the CUSUM and the Kalman under the threshold.

The detection algorithms perform strongly when compared against a constant normal baseline. Having both an accumulative and a residual detector protects against slow drift attacks as well as large anomalous deviations. The random forest complements these well by modelling multiple variable relationships, making masking less effective as it detects deviations in variable relationships. I verified the results of the detectors using multiple normal baselines taken at various times and no false positives or inconsistent results were observed. The stealthy attack consistently took longer to trigger detection than the noisy attacks, meaning it is still detected but at a slower rate. However, during periods of transition none of the detection algorithms can reliably detect the stealthy attack but all three of them correctly flag on the noisy attack. A potential mitigation to this is using variable mapping or limiting high noise time periods.

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